Machine Learning Homework 3

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1 Math Questions

1.1 Question 1

1.

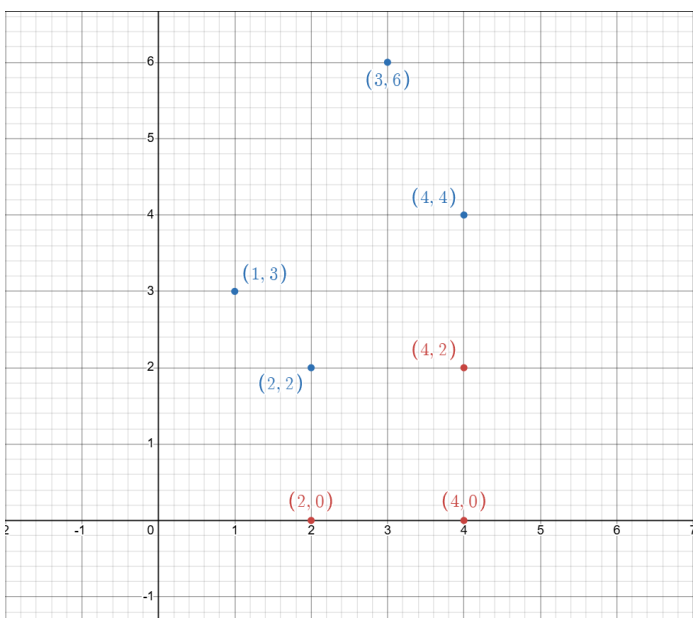

Figure 1: scatter plot

Let $y = +1$ for Blue, and $y = -1$ for Red.

Step 1: Hard-margin constraints.
Find $w = (w_1, w_2)$ and b such that

$$y_i\big(w^\top x_i + b\big) \geq 1, \qquad i = 1, \ldots, 7,$$

and minimize

$$\frac{1}{2}\,\|w\|^2.$$

Support vectors satisfy equality:

$$y_i\big(w^\top x_i + b\big) = 1.$$

In 2D, the parameters w_1, w_2, b are three unknowns, so a candidate solution can be determined by choosing three support vectors (giving three equalities), solving for w and b, and then verifying all inequalities and maximal margin.

Step 2: Pick a candidate set of 3 support vectors.
One typical optimal pattern is to choose two points from one class and one from the other. The points that are "closest across classes" here are:

$$\begin{aligned} \text{Blue 2: } & (2,2), \\ \text{Blue 3: } & (4,4), \\ \text{Red 6: } & (4,2), \\ \text{Red 5: } & (2,0). \end{aligned}$$

We try the situation where two Blue points and one Red point lie on the margins:

$$\begin{aligned} \text{Blue 2 and Blue 3 satisfy } & w^\top x + b = +1, \\ \text{Red 6 satisfies } & w^\top x + b = -1. \end{aligned}$$

So we impose the system:

$$\begin{aligned} w^\top(2,2) + b &= 1 \quad \text{(A)}, \\ w^\top(4,4) + b &= 1 \quad \text{(B)}, \\ w^\top(4,2) + b &= -1 \quad \text{(C)}. \end{aligned}$$

Step 3: Solve the linear system for w_1, w_2, b.
Write out the equations:

$$\begin{aligned} &\text{(A) } 2w_1 + 2w_2 + b = 1, \\ &\text{(B) } 4w_1 + 4w_2 + b = 1, \\ &\text{(C) } 4w_1 + 2w_2 + b = -1. \end{aligned}$$

Subtract (A) from (B):

$$(4w_1-2w_1)+(4w_2-2w_2)=0 \implies 2w_1+2w_2=0 \implies w_1+w_2=0 \implies w_2=-w_1. \tag{1}$$

Now subtract (A) from (C):

$$(4w_1-2w_1)+(2w_2-2w_2)=-2 \implies 2w_1=-2 \implies w_1=-1 \implies w_2=1.$$

Plug into (A):

$$2(-1)+2(1)+b=1 \implies 0+b=1 \implies b=1.$$

So a candidate solution is:

$$w=(-1,\,1), \qquad b=1.$$

The corresponding hyperplane is:

$$w^\top x+b=0 \implies -x_1+x_2+1=0 \implies x_2=x_1-1.$$

Step 4: Verify all constraints.
We must check $y_i(w^\top x_i+b)\geq 1$ for all points.
Compute

$$f(x)=w^\top x+b=-x_1+x_2+1.$$

Blue points ($y=+1$): need $f(x)\geq 1$

$$\begin{aligned}
(3,6): f&=-3+6+1=4\ \checkmark\\
(2,2): f&=-2+2+1=1\ \checkmark\ \text{(SV)}\\
(4,4): f&=-4+4+1=1\ \checkmark\ \text{(SV)}\\
(1,3): f&=-1+3+1=3\ \checkmark
\end{aligned}$$

Red points ($y=-1$): need $f(x)\leq -1$

$$\begin{aligned}
(2,0): f&=-2+0+1=-1\ \checkmark\ \text{(SV)}\\
(4,2): f&=-4+2+1=-1\ \checkmark\ \text{(SV)}\\
(4,0): f&=-4+0+1=-3\ \checkmark
\end{aligned}$$

All constraints are satisfied; we have a feasible separator with multiple active (equality) constraints.

Maximal-margin hyperplane:

$$-x_1+x_2+1=0 \quad \text{(equivalently } x_1-x_2-1=0\text{)}.$$

This is seen on figure 2 below.

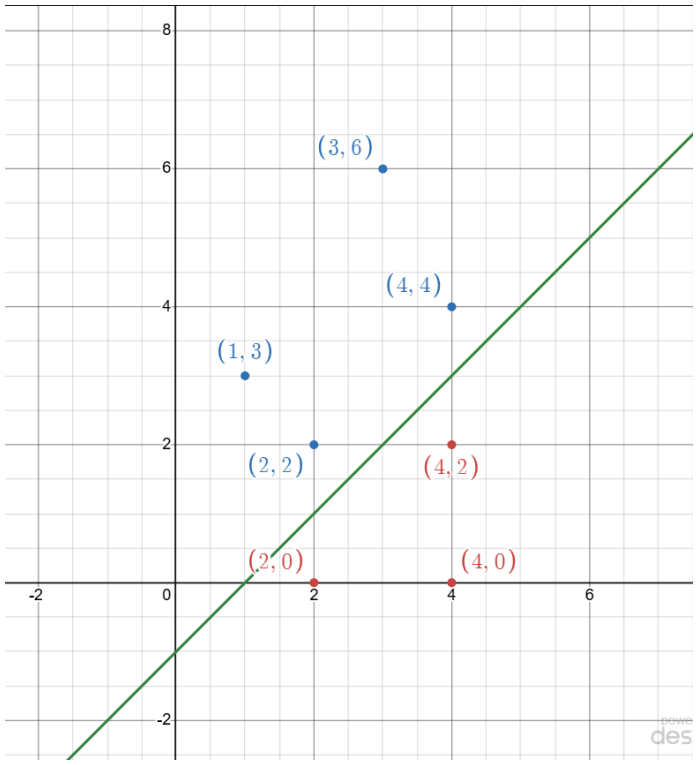

Figure 2: scatter plot with Linear SVM

2.

From Q1, the maximal-margin separating hyperplane is:

$$x_1 - x_2 - 1 = 0.$$

We write the classifier in the standard form:

$$f(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

and classify by the sign of $f(x)$.
So,

$$f(x) = x_1 - x_2 - 1.$$

Therefore, the parameters are

$$\beta_0 = -1, \qquad \beta_1 = 1, \qquad \beta_2 = -1.$$

Classification rule

$$\hat{Y} = \text{Blue} \quad \text{if } f(x) < 0 \text{ (i.e., } x_2 > x_1 - 1),$$

$$\hat{Y} = \text{Red} \quad \text{if } f(x) > 0 \text{ (i.e., } x_2 < x_1 - 1).$$

3.

From Q1/Q2 the separating hyperplane is

$$x_1 - x_2 - 1 = 0,$$

so

$$w = (\beta_1, \beta_2) = (1, -1).$$

For the maximal-margin (canonical) hyperplane, where support vectors satisfy

$$y_i\big(w^\top x_i + b\big) = 1,$$

the margin (distance from the hyperplane to the closest point) is

$$\text{margin} = \frac{1}{\|w\|}.$$

Compute:

$$\|w\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}.$$

So the margin is

$$\frac{1}{\sqrt{2}}.$$

The total width of the margin band between the two margin lines is

$$\frac{2}{\|w\|} = \frac{2}{\sqrt{2}} = \sqrt{2}.$$

4.

From Q1, the support vectors are the points $(2, 2)$ and $(4, 4)$ for Blue, and $(2, 0)$ and $(4, 2)$ for Red.

5. Observation 7 is the Red point (4,0). Point (4,0) is not a support vector. So, a slight movement of this point will not affect the maximal margin hyperplane.

6.

A non-optimal separating hyperplane can be obtained by slightly shifting the intercept β_0 For example:

The line

$$x_1 - x_2 - 0.5 = 0 \quad \text{(equivalently } x_2 = x_1 - 0.5\text{)}$$

is our candidate separator.

Check separation using $g(x) = x_1 - x_2 - 0.5$, predicting Red if $g(x) > 0$ and Blue if $g(x) < 0$.

Blue points:

$$\begin{aligned}
(3,6) &: 3 - 6 - 0.5 = -3.5 < 0, \\
(2,2) &: 2 - 2 - 0.5 = -0.5 < 0, \\
(4,4) &: 4 - 4 - 0.5 = -0.5 < 0, \\
(1,3) &: 1 - 3 - 0.5 = -2.5 < 0.
\end{aligned}$$

Red points:

$$\begin{aligned}
(2,0) &: 2 - 0 - 0.5 = 1.5 > 0, \\
(4,2) &: 4 - 2 - 0.5 = 1.5 > 0, \\
(4,0) &: 4 - 0 - 0.5 = 3.5 > 0.
\end{aligned}$$

It separates, but it's not the maximal-margin separator.

7.

To make the classes not linearly separable, we need to add a misclassified point. The new Point (2,3) with Color Red will make the classes linearly inseparable. Check figure 3 below.

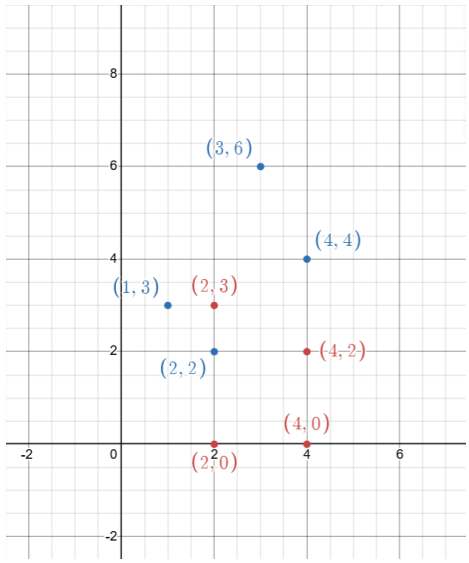

Figure 3: Linearly inseparability

1.2 Question 2

1.

Given data

Positive class $(+1)$: $(1, 1)$, $(-1, -1)$

Negative class (-1): $(1, -1)$, $(-1, 1)$

A table (rows are samples):

i	x_1	x_2	y
1	1	1	$+1$
2	-1	-1	$+1$
3	1	-1	-1
4	-1	1	-1

So:

$$X \in \mathbb{R}^{4\times 2} \quad \text{(shape (4,2))}$$

$$y \in \mathbb{R}^4 \quad \text{(or shape (4,1))}$$

Bonus (logic gate): This dataset corresponds to the XNOR (equivalence) function: output $+1$ when the inputs have the same sign, and -1 when they differ.

2. For illustration purposes, let points with y = +1 be Blue and points with y = -1 be Red. Check figure 4 below.

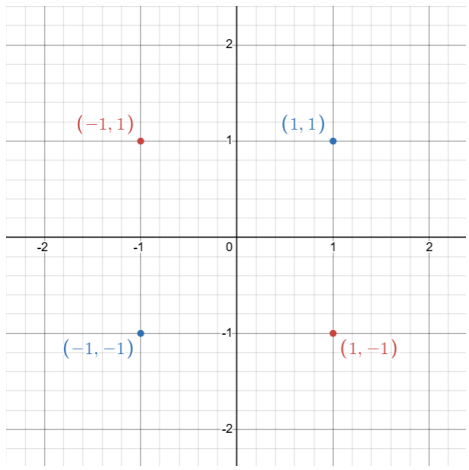

Figure 4: XNOR Scatter Plot

From figure 4, we see that data is not linearly separable.

3.

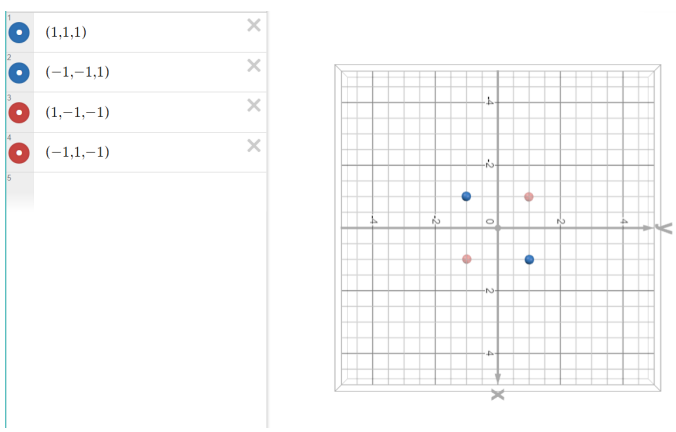

Figure 5: 3D View 1

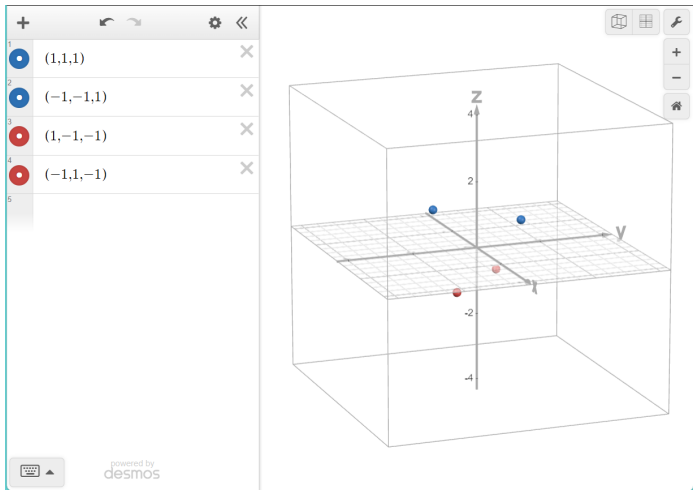

Figure 6: 3D View 2

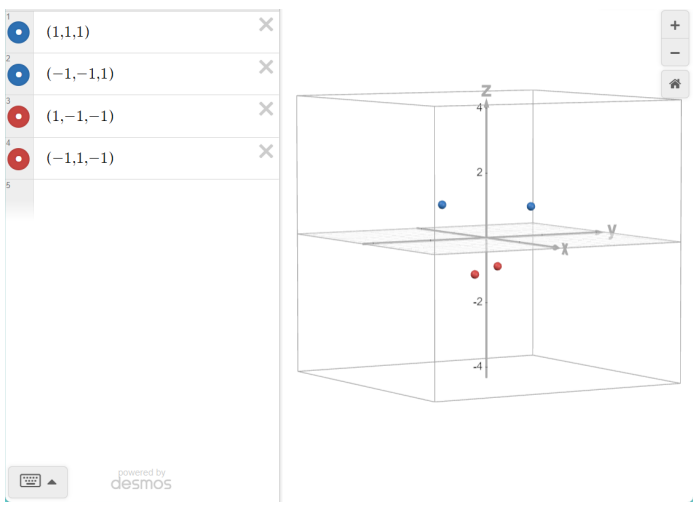

Figure 7: 3D view 3

From the 3D views above, we see that increasing the dimension of data through feature transformation makes the data linearly separable.

4. We transform each input $x = (x_1, x_2)$ to

$$\phi(x) = z = (z_1, z_2, z_3) = (x_1,\ x_2,\ x_1x_2).$$

The transformed training set is:
Positive ($y = +1$):

$$z^{(1)} = (1, 1, 1), \qquad z^{(2)} = (-1, -1, 1)$$

Negative ($y = -1$):

$$z^{(3)} = (1, -1, -1), \qquad z^{(4)} = (-1, 1, -1)$$

1) Hard-margin SVM in z-space
Find $w \in \mathbb{R}^3$, $b \in \mathbb{R}$ minimizing

$$\frac{1}{2}\|w\|^2$$

subject to

$$y_i\big(w^\top z^{(i)} + b\big) \geq 1, \qquad i = 1, \ldots, 4.$$

Support vectors satisfy

$$y_i\big(w^\top z^{(i)} + b\big) = 1.$$

2) Use symmetry to find the optimal separating hyperplane
Observe:

$$z_3 = +1 \text{ for all positive points}, \qquad z_3 = -1 \text{ for all negative points}.$$

Thus a hyperplane depending only on z_3 separates perfectly. Consider

$$w = (0, 0, \alpha), \qquad b = 0.$$

Then

$$w^\top z + b = \alpha z_3.$$

Constraints:
Positive ($y = +1, z_3 = +1$):

$$+1(\alpha \cdot 1) \geq 1 \ \Rightarrow\ \alpha \geq 1.$$

Negative ($y = -1, z_3 = -1$):

$$-1(\alpha \cdot (-1)) \geq 1 \ \Rightarrow\ \alpha \geq 1.$$

Objective:

$$\frac{1}{2}\|w\|^2 = \frac{1}{2}\alpha^2.$$

Minimized by the smallest feasible α, so

$$\alpha^* = 1.$$

Thus the optimal SVM parameters are:

$$w^* = (0, 0, 1), \qquad b^* = 0.$$

Decision boundary:

$$w^{*\top} z + b^* = z_3 = 0.$$

3) Margin

For a canonical hard-margin SVM:

$$\gamma = \frac{1}{\|w^*\|}.$$

Here:

$$\|w^*\| = \sqrt{0^2 + 0^2 + 1^2} = 1,$$

so

$$\gamma = 1.$$

Full margin band width:

$$2\gamma = 2.$$

4) Support vectors

Functional margins:

$$m_i = y_i\big(w^{*\top} z^{(i)} + b^*\big) = y_i z_3^{(i)}.$$

Since $y = \pm 1$ and $z_3 = \pm 1$:
Positive: $y = +1, z_3 = +1 \Rightarrow m = 1$
Negative: $y = -1, z_3 = -1 \Rightarrow m = 1$
Thus every point satisfies $m_i = 1$, meaning:

All 4 points are support vectors.

Final answers for Q2.4

Optimal hyperplane in transformed space:

$$z_3 = 0$$

with

$$w = (0, 0, 1), \qquad b = 0.$$

Margin size:

$$1.$$

Support vectors:

$$\text{All four transformed samples.}$$

Check Figure 8 below. We see that $z_3 = 0$, represented by the green plane, successfully separates the data linearly.

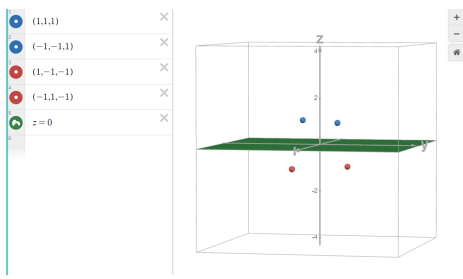

Figure 8: 3D Linearly Separable

2 Programming Questions

2.1 Question 3

2.1.1 (a) Data Pre-processing

1. Done
2.
Done
3.
Done

2.1.2 (b) Model Initialization

1. Done
2. Done
3. Done

2.1.3 (c) Feature Selection and Model Training

Done

2.1.4 (d) Support Vectors

Below I show the first 10 support vectors for each trained model:

```
Support vectors for ('linear', ('CGPA', 'SOP')):
[8.54 3.5 ]
[8.7 4. ]
[8.53 3.5 ]
[8.2 4. ]
[8.49 3.  ]
[8.68 4.  ]
[9.  2.5]
[8.67 4.  ]
[8.84 3.  ]
[8.47 4.5 ]
```

Figure 9: 'linear', ('CGPA', 'SOP')

```
Support vectors for ('rbf', ('CGPA', 'SOP')):
[8.54 3.5 ]
[8.7 4. ]
[8.53 3.5 ]
[8.2 4. ]
[8.49 3.  ]
[8.68 4.  ]
[9.  2.5]
[8.67 4.  ]
[8.84 3.  ]
[8.47 4.5 ]
```

Figure 10: 'rbf', ('CGPA', 'SOP')

```
Support vectors for ('poly', ('CGPA', 'SOP')):
[8.54 3.5 ]
[8.7 4. ]
[8.53 3.5 ]
[8.49 3.  ]
[8.68 4.  ]
[9.  2.5]
[8.67 4.  ]
[8.84 3.  ]
[8.47 4.5 ]
[8.73 3.  ]
```

Figure 11: 'poly', ('CGPA', 'SOP')

```
Support vectors for ('linear', ('CGPA', 'GRE Score')):
[  8.7 317. ]
[  8.53 312.  ]
[  8.68 321.  ]
[  9. 316.]
[  8. 322.]
[  8.67 318.  ]
[  8.84 327.  ]
[  8.47 317.  ]
[  8.73 316.  ]
[  8.46 315.  ]
```

Figure 12: 'linear', ('CGPA', 'GRE Score')

```
Support vectors for ('rbf', ('CGPA', 'GRE Score')):
[  8.27 312.  ]
[  8.12 309.  ]
[  8.7 317. ]
[  8.53 312.  ]
[  8.2 307. ]
[  8.49 308.  ]
[  8.68 321.  ]
[  9. 316.]
[  8. 322.]
[  8.67 318.  ]
```

Figure 13: 'rbf', ('CGPA', 'GRE Score')

```
Support vectors for ('poly', ('CGPA', 'GRE Score')):
[  8.27 312.  ]
[  8.7 317. ]
[  8.53 312.  ]
[  8.68 321.  ]
[  9. 316.]
[  8. 322.]
[  8.67 318.  ]
[  8.01 312.  ]
[  8.84 327.  ]
[  8.47 317.  ]
```

Figure 14: 'poly', ('CGPA', 'GRE Score')

```
Support vectors for ('linear', ('SOP', 'LOR')):
[3.5 4. ]
[4. 3.]
[3.5 3. ]
[4. 3.]
[3.  3.5]
[4. 4.]
[3.  3.5]
[4.  3.5]
[3.  3.5]
[3.5 4. ]
```

Figure 15: 'linear', ('SOP', 'LOR')

```
Support vectors for ('rbf', ('SOP', 'LOR')):
[3.5 4. ]
[4. 3.]
[3.5 3. ]
[4. 3.]
[4. 4.]
[4.  3.5]
[1.  1.5]
[3.5 4. ]
[4.5 4. ]
[3.5 4. ]
```

Figure 16: ('rbf', ('SOP', 'LOR')

```
Support vectors for ('poly', ('SOP', 'LOR')):
[3.5 4. ]
[4. 3.]
[3.5 3. ]
[4. 3.]
[3.  3.5]
[4. 4.]
[3.  3.5]
[4.  3.5]
[3.  3.5]
[3.5 4. ]
```

Figure 17: 'poly', ('SOP', 'LOR')

```
Support vectors for ('linear', ('LOR', 'GRE Score')):
[  3.5 312. ]
[  3. 317.]
[  4. 321.]
[  3. 316.]
[  3.5 322. ]
[  3.5 318. ]
[  3.5 327. ]
[  4. 310.]
[  4. 317.]
[  3.5 316. ]
```

Figure 18: 'linear', ('LOR', 'GRE Score')

```
Support vectors for ('rbf', ('LOR', 'GRE Score')):
[  3.5 312. ]
[  3. 309.]
[  3. 317.]
[  3. 312.]
[  3. 307.]
[  3.5 308. ]
[  4. 321.]
[  3. 316.]
[  3.5 322. ]
[  3.5 318. ]
```

Figure 19: 'rbf', ('LOR', 'GRE Score')

```
Support vectors for ('poly', ('LOR', 'GRE Score')):
[  3.5 312. ]
[  3. 317.]
[  3. 312.]
[  4. 321.]
[  3. 316.]
[  3.5 322. ]
[  3.5 318. ]
[  1.5 312. ]
[  3.5 327. ]
[  4. 317.]
```

Figure 20: 'poly', ('LOR', 'GRE Score')

2.1.5 (e) Result Visualization

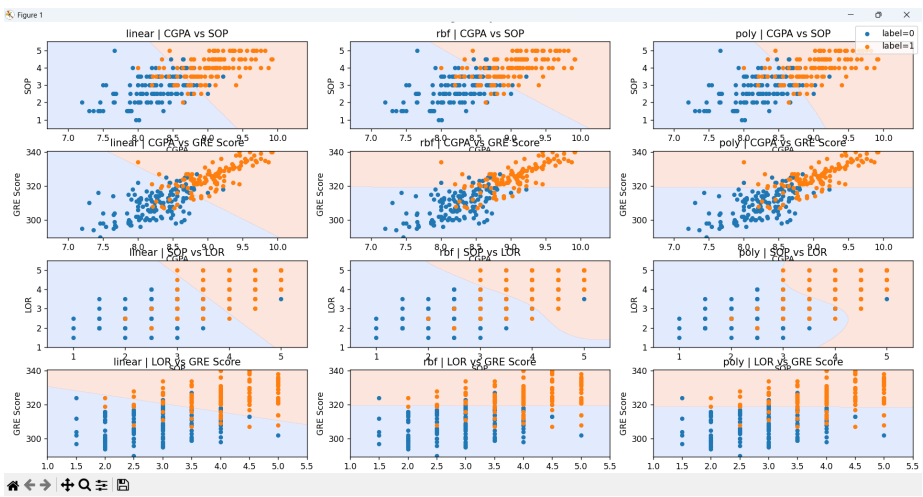

Figure 21: Result Visualization

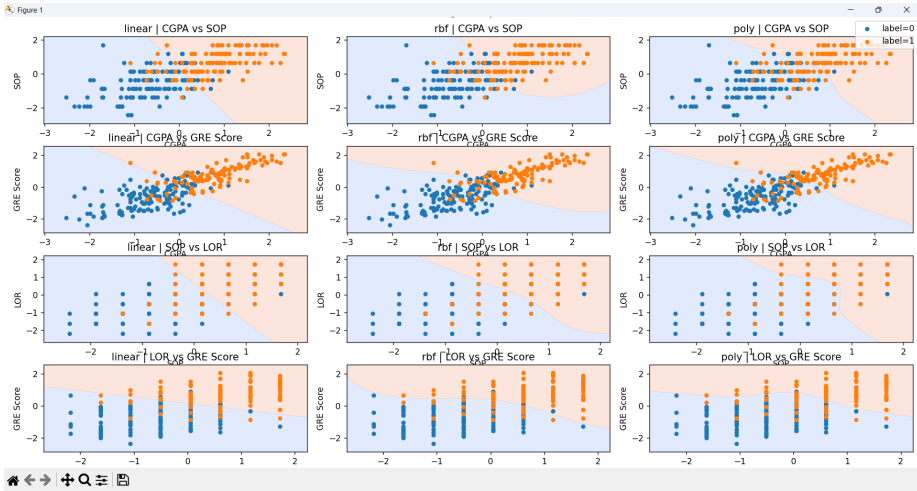

Figure 22: Result Visualization with Standardized Data

2.1.6 (f) Result Analysis

When I standardize the data, my_best_model is

$$\text{SVC}(\text{kernel} = \text{"linear"},\ C = 1.0),$$

which fails test `3.6) test_support_vectors`.
When I do *not* standardize the data, my_best_model is

$$\text{SVC}(\text{kernel} = \text{"rbf"},\ C = 10.0,\ gamma = 0.5),$$

which passes the test.
Therefore, we should not standardize the data. Thus,

$$\text{my_best_model} = \text{SVC}(\text{kernel} = \text{"rbf"},\ C = 10.0,\ gamma = 0.5).$$

3 3 Use of AI and External Resources

1. What were the prompts you provided the AI tools?

a) Explain why dropping Chance of Admit is necessary.

b) Gradescope says: 'Wrong number of support vectors for linear kernel (31 != 34). What could cause this and what code section should I adjust?

c) Provide a mathematical procedure to find the maximal-margin hyperplane and identify support vectors for the given 2D labeled points.

2. What mistakes did you catch the AI making?

a) The AI initially suggested the mismatch in the number of support vectors was mainly due to shuffling/seed/hyperparameters. After testing, I found the real cause for the autograder's strict "support vector count" check was feature standardization.

3. What did you do to validate the correctness of the AI generated portions, aside from relying on Gradescope Autograder scoring?

a) Verified label is recomputed from the median of Chance of Admit

b) Confirmed training/validation split happens before any operations that could leak information.